STUDENT PERFORMANCE ANALYSIS AND PREDICTION SYSTEM

MINOR PROJECT REPORT

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ABSTRACT

The Student Performance Analysis and Prediction System is a Python-based machine learning project developed to analyze students' academic performance and predict their expected final examination marks.

The project uses historical student data to identify relationships between academic factors such as study time, previous examination marks, and final examination scores.

The system involves data collection, preprocessing, exploratory data analysis, feature engineering, machine learning model development, and performance evaluation.

Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn are used to perform data analysis, generate visualizations, and develop a Linear Regression model.

The model predicts students' final examination marks based on selected academic features. Its performance is evaluated using Mean Absolute Error, Root Mean Squared Error, and the R^2 score.

The project aims to identify academic performance patterns and help recognize students who may benefit from additional academic support.

Keywords: Student Performance, Machine Learning, Linear Regression, Data Analysis, Python, Academic Prediction.

1. INTRODUCTION

Education plays an important role in the development of individuals and society. Understanding students' academic performance helps educational institutions identify learning difficulties and provide appropriate support.

Traditional methods of analyzing academic performance mainly depend on examination marks and manual calculations. These methods may not provide sufficient information about the factors associated with student performance.

Machine Learning provides methods to analyze historical academic data and identify patterns that may help predict future examination results.

The Student Performance Analysis and Prediction System uses Python and machine learning techniques to analyze student data and predict expected examination marks.

The system considers relevant academic features such as study time, previous examination marks, and absences to estimate final examination scores.

The project also uses data visualization techniques to present academic trends and relationships in an understandable manner.

2. PROBLEM STATEMENT

Students' academic performance can be influenced by several factors, including study habits, previous examination results, and attendance.

Analyzing these factors manually can be time-consuming and may not reveal important relationships within the data.

Therefore, a machine learning-based system is required to analyze historical student data, identify academic performance patterns, and predict expected final examination marks.

The proposed system uses Linear Regression to predict students' final examination scores and provides evaluation results to understand the model's performance.

3. PROJECT OBJECTIVES

The main objectives of the project are:

1. To collect and analyze student academic performance data.
2. To identify and handle missing values and duplicate records.
3. To perform data cleaning and feature engineering.
4. To calculate descriptive statistics such as mean, median, and standard deviation.
5. To visualize academic performance using different charts.
6. To analyze relationships between study time, previous marks, and final examination scores.
7. To develop a Linear Regression model for predicting final examination marks.
8. To evaluate the model using appropriate regression metrics.
9. To identify academic performance patterns and students who may benefit from additional support.
10. To document the findings, limitations, and future improvements.

4. SCOPE OF THE PROJECT

The project focuses on analyzing student academic performance using historical student data.

The system includes:

- Dataset collection and preprocessing.
- Exploratory Data Analysis.
- Feature engineering.
- Academic performance visualization.
- Linear Regression model development.
- Model evaluation.
- Prediction of final examination marks.
- Interpretation of academic performance patterns.

The project is intended for academic analysis and educational support. It does not replace teachers' professional judgment or independently determine a student's academic ability.

5. DATASET DESCRIPTION

5.1 Dataset Source

The project uses the Student Performance dataset available from the UCI Machine Learning Repository.

Dataset source:

https://archive.ics.uci.edu/dataset/320/student+performance

The dataset contains information about students' academic performance and relevant personal, social, and school-related factors.

The mathematics student dataset is used for this project.

5.2 Dataset Features

The dataset contains several attributes related to students' academic performance.

Feature	Description
studytime	Weekly study time category
G1	First-period examination grade
G2	Second-period examination grade
G3	Final examination grade
absences	Number of school absences

The project selects study time, previous examination grades, and absences as input features.

The final examination grade, represented by G3, is used as the target variable.

5.3 Dataset Size

Number of records: [395]

Number of columns: [33]

Number of input features used: 4

Target variable: G3 — Final Examination Grade

```
[5]: df = pd.read_csv(DATA_PATH, sep=";")

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)

df.head()
```

```
Dataset loaded successfully.
Dataset shape: (395, 33)
```

[5]:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	hea
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	

5 rows × 33 columns

6. TECHNOLOGIES AND LIBRARIES USED

6.1 Python

Python is used as the main programming language because of its simplicity and availability of machine learning and data analysis libraries.

6.2 NumPy

NumPy is used for numerical operations and mathematical calculations.

6.3 Pandas

Pandas is used to load, inspect, clean, and manipulate the student dataset.

6.4 Matplotlib

Matplotlib is used to generate charts and visualize academic performance patterns.

6.5 Seaborn

Seaborn is used to create statistical visualizations, including box plots and correlation heatmaps.

6.6 Scikit-learn

Scikit-learn is used to split the dataset, train the Linear Regression model, perform cross-validation, and evaluate model performance.

6.7 Jupyter Notebook

Jupyter Notebook is used to implement the project and display the code, results, and visualizations.

7. DATA PREPROCESSING AND FEATURE ENGINEERING

Data preprocessing is an important step in preparing the dataset for machine learning.

7.1 Data Loading

The student dataset is loaded using the Pandas library.

The dataset is inspected to understand its structure, column names, data types, and number of records.

7.2 Handling Missing Values

Missing values are identified using Pandas.

Records with missing target values are removed because the model requires actual final examination marks during training.

Missing values in input features are handled using median imputation within the machine learning pipeline.

7.3 Duplicate Removal

Duplicate records are identified and removed to avoid unnecessary repetition in the dataset.

[6]:

```
print("Column names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()

print("\nMissing values:")
print(df.isnull().sum())

print("\nDuplicate records:")
print(df.duplicated().sum())

print("\nFirst five records:")
display(df.head())
```

```
Column names:
['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu', 'Mjob', 'Fjob', 'reason', 'guardian', 't
raveltime', 'studytime', 'failures', 'schoolsup', 'famsup', 'paid', 'activities', 'nursery', 'higher', 'internet',
'romantic', 'famrel', 'freetime', 'goout', 'Dalc', 'Walc', 'health', 'absences', 'G1', 'G2', 'G3']

Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    str
 1   sex         395 non-null    str
 2   age         395 non-null    int64
 3   address     395 non-null    str
 4   famsize     395 non-null    str
 5   Pstatus     395 non-null    str
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    str
 9   Fjob        395 non-null    str
 10  reason      395 non-null    str
 11  guardian    395 non-null    str
 12  traveltime  395 non-null    int64
 13  studytime   395 non-null    int64
 14  failures    395 non-null    int64
 15  schoolsup   395 non-null    str
 16  famsup      395 non-null    str
 17  paid        395 non-null    str
 18  activities  395 non-null    str
 19  nursery     395 non-null    str
```

```
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64

Duplicate records:
0

First five records:
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	hea
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	

5 rows × 33 columns

7.4 Feature Selection

The following features are selected as inputs:

- Study time.
- First-period examination marks.
- Second-period examination marks.
- Number of absences.

The final examination marks are selected as the target variable.

7.5 Feature Engineering

The selected academic features are assigned meaningful names for analysis and interpretation.

The target variable is also used to identify students who passed or failed based on a passing score of 10 out of 20.

The pass/fail feature is used for analysis only and is not required for the Linear Regression model.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis is performed to understand the dataset and identify important academic performance patterns.

The analysis includes descriptive statistics and visualizations.

8.1 Descriptive Statistics

The following statistical measures are calculated:

Mean: The average value of a numerical feature.

Median: The middle value of a numerical feature after arranging the values in order.

Standard Deviation: A measure of how much the values vary from the mean.

These measures help understand the distribution of student examination marks and other academic features.

[9]:
```
print("Mean:")
display(df.mean(numeric_only=True))

print("\nMedian:")
display(df.median(numeric_only=True))

print("\nStandard deviation:")
display(df.std(numeric_only=True))

print("\nSummary statistics:")
display(df.describe())
```

```
Mean:
age               16.696203
Medu               2.749367
Fedu               2.521519
traveltime         1.448101
studytime          2.035443
failures           0.334177
famrel             3.944304
freetime           3.235443
goout              3.108861
Dalc               1.481013
Walc               2.291139
health             3.554430
absences           5.708861
G1                10.908861
G2                10.713924
G3                10.415190
StudyHours         2.035443
PreviousMarks     10.908861
FinalMarks        10.415190
Absences           5.708861
Passed             0.670886
dtype: float64

Median:
age               17.0
Medu               3.0
Fedu               2.0
traveltime         1.0
studytime          2.0
failures           0.0
famrel             4.0
freetime           3.0
goout              3.0
```

```
Standard deviation:
age               1.276043
Medu              1.094735
Fedu              1.088201
traveltime        0.697505
studytime         0.839240
failures          0.743651
famrel            0.896659
freetime          0.998862
goout             1.113278
Dalc              0.890741
Walc              1.287897
health            1.390303
absences          8.003096
G1                3.319195
G2                3.761505
G3                4.581443
StudyHours        0.839240
PreviousMarks     3.319195
FinalMarks        4.581443
Absences          8.003096
Passed            0.470487
dtype: float64
```

Summary statistics:

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443	3.108861	1.481013
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862	1.113278	0.890741
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000	2.000000	1.000000
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000	3.000000	1.000000
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000	4.000000	2.000000
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000	5.000000	5.000000

8 rows × 21 columns

8.2 Bar Chart

A bar chart is created to compare average final examination marks across different study time categories.

This visualization helps examine whether students with different study time categories have different average examination scores.

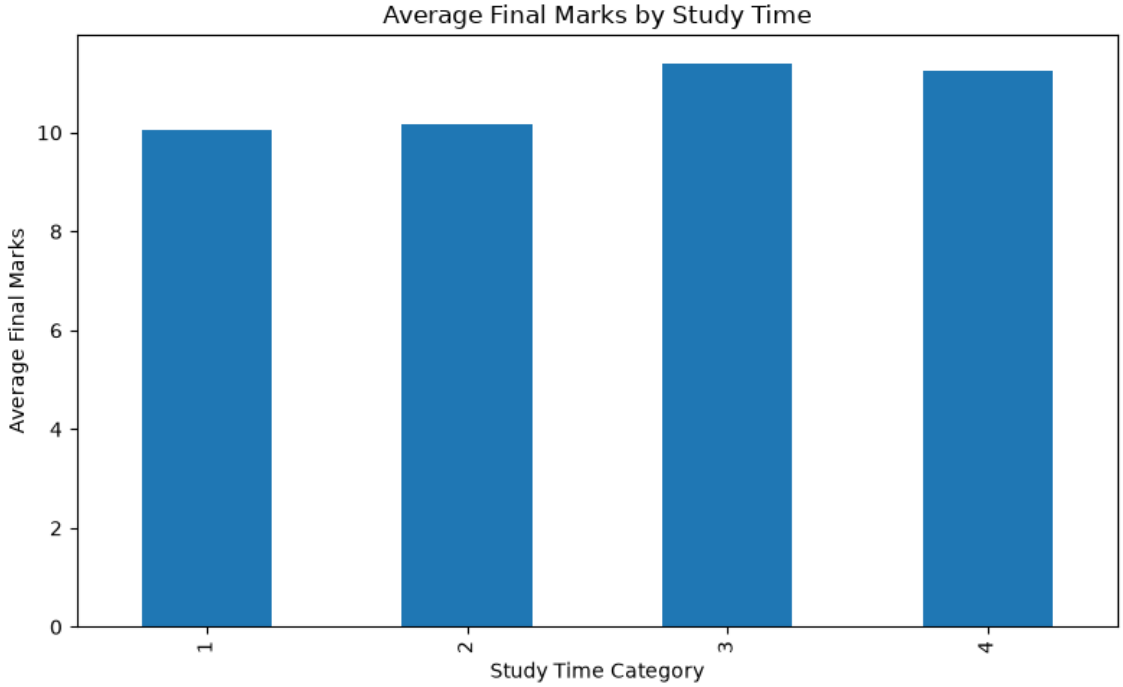
[12]:
```
plt.figure(figsize=(8, 5))

df.groupby("studytime")["G3"].mean().plot(
    kind="bar"
)

plt.title("Average Final Marks by Study Time")
plt.xlabel("Study Time Category")
plt.ylabel("Average Final Marks")
plt.tight_layout()

plt.savefig(
    os.path.join(
        VISUALIZATION_DIR,
        "study_time_bar_chart.png"
    )
)

plt.show()
```



8.3 Histogram

A histogram is generated to display the distribution of final examination marks.

It helps identify the range of marks and the concentration of students within different score intervals.

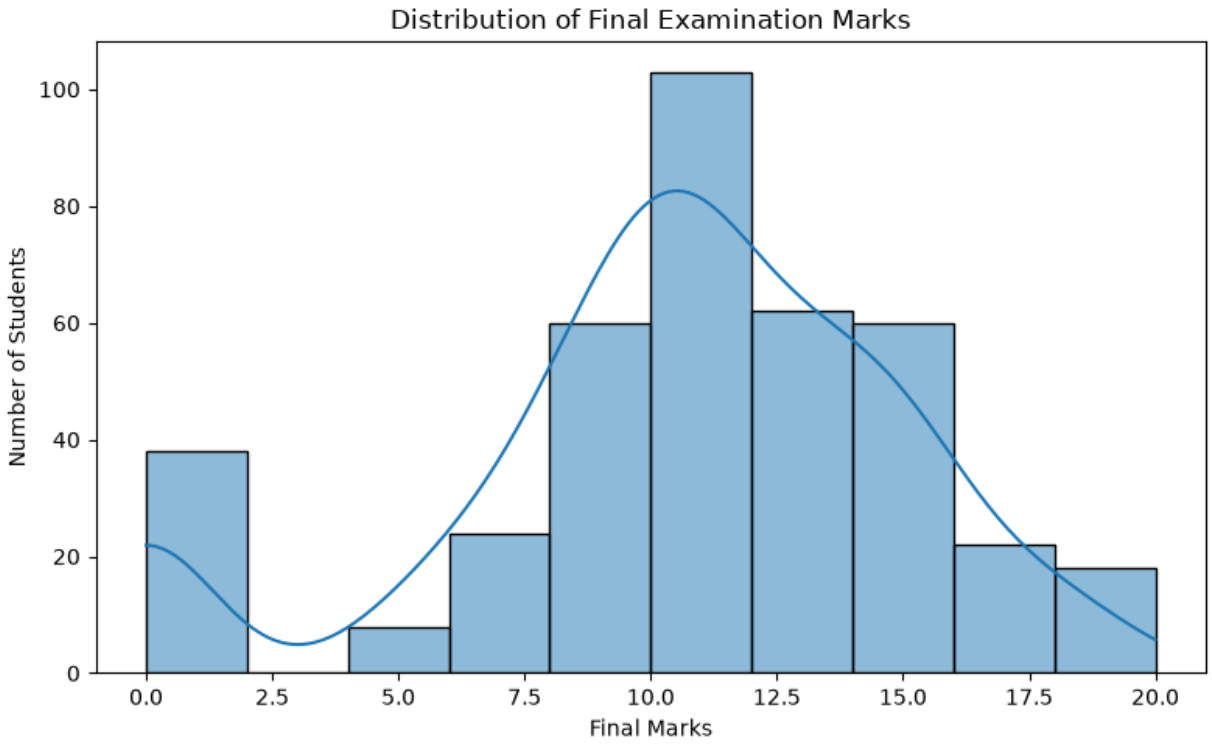
```
[13]: plt.figure(figsize=(8, 5))

sns.histplot(
    df["FinalMarks"],
    bins=10,
    kde=True
)

plt.title("Distribution of Final Examination Marks")
plt.xlabel("Final Marks")
plt.ylabel("Number of Students")
plt.tight_layout()

plt.savefig(
    os.path.join(
        VISUALIZATION_DIR,
        "final_marks_histogram.png"
    )
)

plt.show()
```



8.4 Scatter Plot

A scatter plot is created to examine the relationship between study time and final examination marks.

It helps visualize whether changes in study time categories are associated with changes in final examination scores.

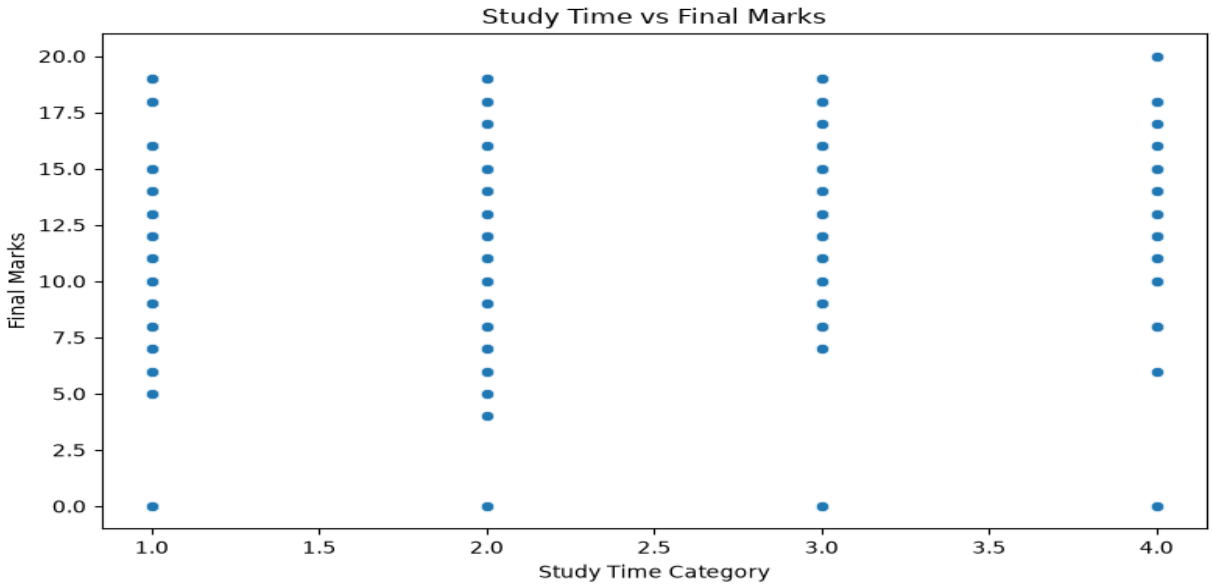
[14]:
```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x="StudyHours",
    y="FinalMarks"
)

plt.title("Study Time vs Final Marks")
plt.xlabel("Study Time Category")
plt.ylabel("Final Marks")
plt.tight_layout()

plt.savefig(
    os.path.join(
        VISUALIZATION_DIR,
        "study_time_scatter_plot.png"
    )
)

plt.show()
```



8.5 Box Plot

A box plot is generated to compare the distribution of final examination marks across study time categories.

It helps identify the median, spread, and possible outliers in examination scores.

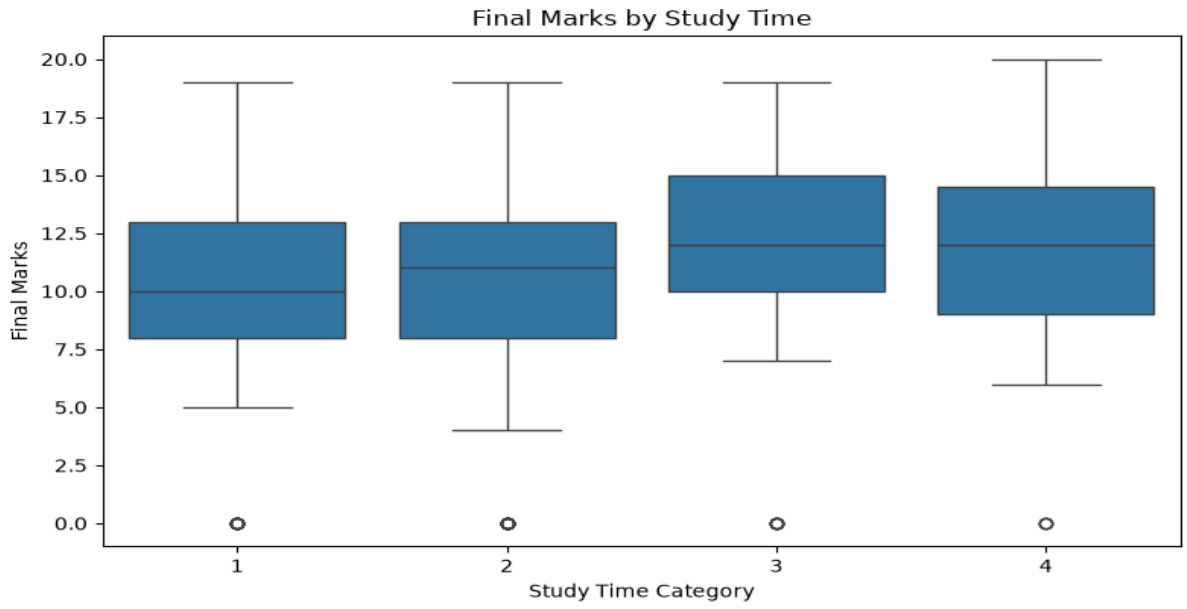
[15]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="studytime",
    y="FinalMarks"
)

plt.title("Final Marks by Study Time")
plt.xlabel("Study Time Category")
plt.ylabel("Final Marks")
plt.tight_layout()

plt.savefig(
    os.path.join(
        VISUALIZATION_DIR,
        "study_time_box_plot.png"
    )
)

plt.show()
```



8.6 Correlation Heatmap

A correlation heatmap is generated to examine the relationships between numerical academic features.

It helps identify features that have stronger positive or negative relationships with final examination marks.

```
[16]: numeric_df = df.select_dtypes(
          include=["number"]
      )

      correlation = numeric_df.corr()

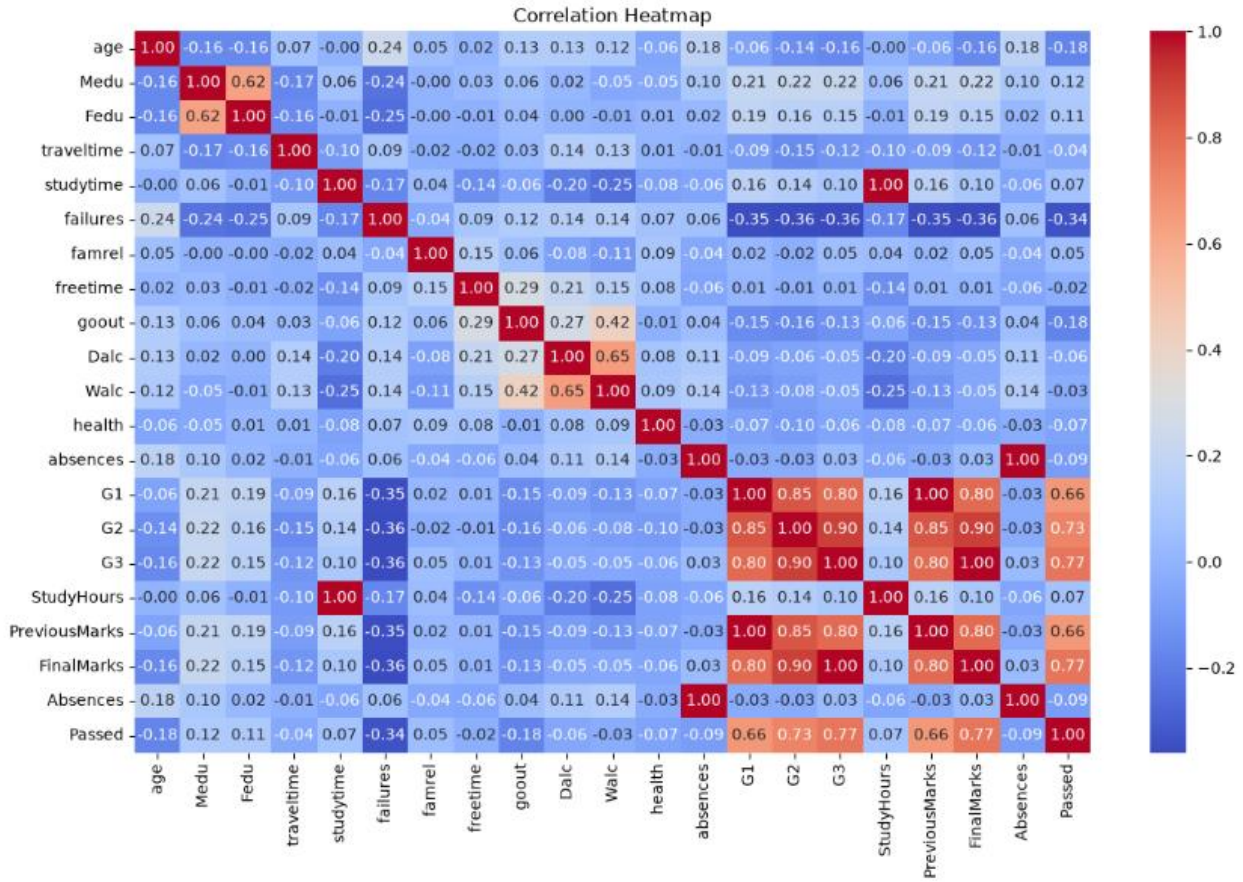
      plt.figure(figsize=(12, 8))

      sns.heatmap(
          correlation,
          annot=True,
          cmap="coolwarm",
          fmt=".2f"
      )

      plt.title("Correlation Heatmap")
      plt.tight_layout()

      plt.savefig(
          os.path.join(
              VISUALIZATION_DIR,
              "correlation_heatmap.png"
          )
      )

      plt.show()
```



8.7 EDA Findings

The following observations are recorded after analyzing the generated visualizations:

1. The relationship between study time and final examination marks: [Students with more study time generally tend to achieve slightly higher final marks.].
2. The relationship between previous examination marks and final examination marks: [Previous examination marks (G1 and G2) show a strong positive relationship with final marks (G3).].
3. The relationship between absences and final examination marks: [Higher absences generally show a slight negative relationship with final marks.].
4. The overall distribution of final examination marks: Final marks are mostly concentrated in the middle range, with fewer students scoring very low or very high marks.]

9. MACHINE LEARNING METHODOLOGY

9.1 Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values.

In this project, Linear Regression is used to predict students' final examination marks.

The model learns the relationship between the selected academic features and the final examination scores using historical student data.

The general Linear Regression equation is:

$$Y = b0 + b1X1 + b2X2 + b3X3 + b4X4$$

Where:

- Y represents the predicted final examination marks.
- b0 represents the intercept.
- X1 represents study time.
- X2 represents first-period examination marks.
- X3 represents second-period examination marks.
- X4 represents the number of absences.

- b1, b2, b3, and b4 represent the model coefficients.

The model estimates the coefficients from the training data and uses them to predict final examination marks for unseen students.

9.2 Input Features

The model uses the following input features:

1. Study time.
2. First-period examination marks.
3. Second-period examination marks.
4. Number of absences.

9.3 Target Variable

The target variable is G3, representing the student's final examination marks.

9.4 Training and Testing Split

The dataset is divided into training and testing sets.

80% of the dataset is used for training.

20% of the dataset is used for testing.

The training data is used to fit the Linear Regression model, while the testing data is used to evaluate its performance on unseen records.

9.5 Model Training

The Linear Regression model is implemented using Scikit-learn.

The model learns the relationship between the input features and final examination marks from the training dataset.

After training, the model generates predictions for the testing dataset.

[19]:
```
model = Pipeline([
    (
        "imputer",
        SimpleImputer(strategy="median")
    ),
    (
        "regressor",
        LinearRegression()
    )
])

model.fit(X_train, y_train)

print("Linear Regression model trained successfully.")
```

```
Linear Regression model trained successfully.
```

10. MODEL EVALUATION

The performance of the Linear Regression model is evaluated using three metrics.

10.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual and predicted examination marks.

A lower MAE indicates that the model's predictions are closer to the actual values.

10.2 Root Mean Squared Error

Root Mean Squared Error measures the square root of the average squared prediction errors.

It gives greater importance to larger prediction errors.

A lower RMSE indicates better prediction performance.

10.3 R^2 Score

The R^2 score measures how well the model explains the variation in the target variable.

A score closer to 1 generally indicates better predictive performance.

10.4 Evaluation Results

Evaluation Metric	Result
Mean Absolute Error	1.31
Root Mean Squared Error	2.04
R^2 Score	0.7962
Average Cross-Validation R^2	0.7962

[21]:
```
mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(
    mean_squared_error(y_test, y_pred)
)

r2 = r2_score(y_test, y_pred)

print("Model Evaluation Results")
print("------------------------")
print("Mean Absolute Error:", round(mae, 2))
print("Root Mean Squared Error:", round(rmse, 2))
print("R2 Score:", round(r2, 4))
```

```
Model Evaluation Results
------------------------
Mean Absolute Error: 1.31
Root Mean Squared Error: 2.04
R2 Score: 0.7962
```

10.5 Cross-Validation

Five-fold cross-validation is performed on the training data.

The training data is divided into five parts, and the model is evaluated across different training and validation combinations.

The average cross-validation R^2 score is used to understand the model's performance across multiple splits.

10.6 Hyperparameter Tuning

Basic hyperparameter tuning is performed using GridSearchCV.

The model's intercept setting is evaluated using different parameter values.

The best parameter configuration is selected based on the cross-validation R^2 score.

11. ACTUAL VS PREDICTED EXAMINATION MARKS

The trained Linear Regression model generates predicted examination marks for the testing dataset.

The actual and predicted marks are compared to understand the model's prediction accuracy.

A scatter plot is generated to visualize the relationship between actual and predicted examination scores.

Points closer to the diagonal reference line indicate predictions closer to the actual marks.

[26]:
```
plt.figure(figsize=(8, 5))

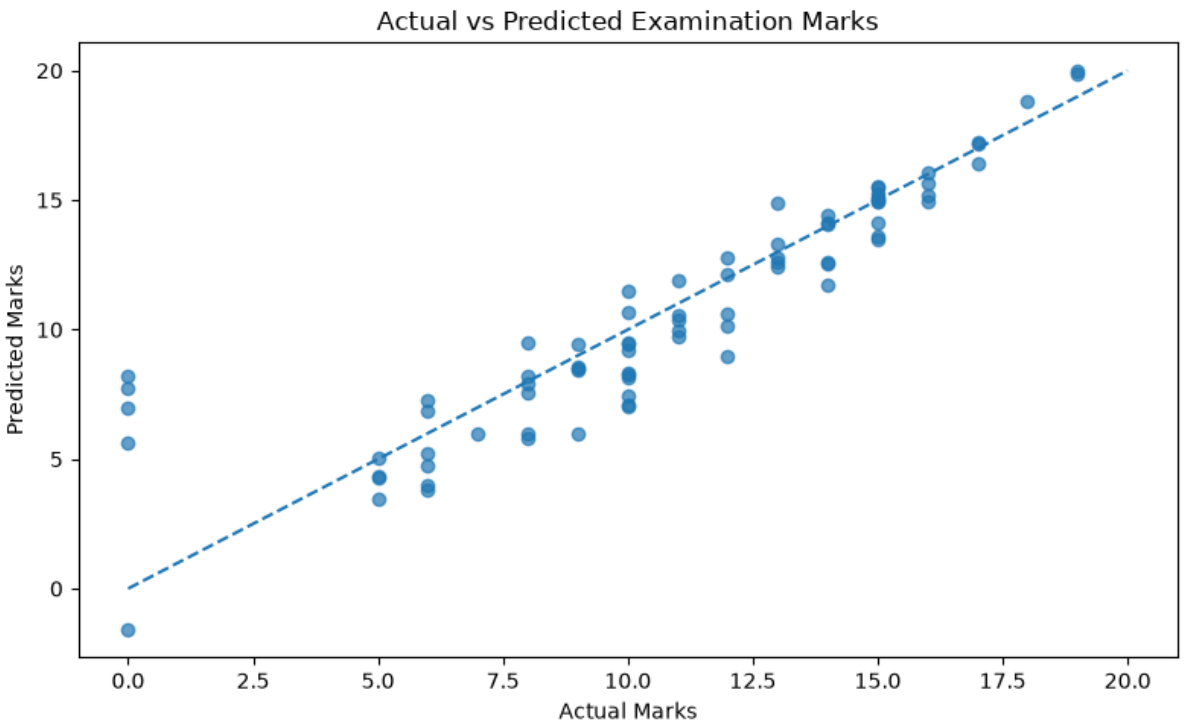
plt.scatter(
    y_test,
    y_pred,
    alpha=0.7
)

plt.plot(
    [0, 20],
    [0, 20],
    linestyle="--"
)

plt.title("Actual vs Predicted Examination Marks")
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.tight_layout()

plt.savefig(
    os.path.join(
        VISUALIZATION_DIR,
        "actual_vs_predicted.png"
    )
)

plt.show()
```



The comparison of actual and predicted marks helps identify the differences between the model's predictions and students' actual examination results.

12. RESULTS AND DISCUSSION

The project successfully implements a machine learning workflow for student performance analysis and prediction.

The dataset is loaded and preprocessed, and exploratory data analysis is performed to understand academic performance patterns.

A Linear Regression model is trained using selected academic features to predict final examination marks.

The model is evaluated using MAE, RMSE, and R^2 score.

The evaluation results are used to understand the model's prediction performance.

The model's actual and predicted examination marks are compared to identify prediction errors.

Results obtained from the implemented model:

- MAE: [Enter actual result].
- RMSE: [Enter actual result].
- R^2 score: [Enter actual result].

The observed relationships between study time, previous examination marks, absences, and final examination scores are documented based on the generated visualizations.

13. LIMITATIONS

The project has the following limitations:

1. The model is trained using a limited student dataset.
2. The dataset may not represent all educational institutions or student populations.
3. The model considers only selected academic features.
4. Factors such as teaching quality, learning environment, and personal circumstances are not fully represented.
5. The model's predictions are estimates and may differ from actual examination results.
6. The model identifies statistical relationships and does not establish that a particular feature causes improved academic performance.

14. FUTURE IMPROVEMENTS

The project can be improved in the following ways:

1. Collect larger and more diverse student datasets.
2. Include additional academic and attendance-related features.
3. Compare Linear Regression with other regression algorithms.
4. Develop an interactive dashboard for students and teachers.
5. Improve the model through additional feature engineering and parameter tuning.
6. Develop a system that can provide personalized academic support recommendations.

15. CONCLUSION

The Student Performance Analysis and Prediction System demonstrates the application of Python and machine learning techniques in educational data analysis.

The project includes dataset preprocessing, exploratory data analysis, feature engineering, Linear Regression model development, and model evaluation.

The system predicts students' final examination marks using selected academic features and provides evaluation results to understand its performance.

The generated visualizations help identify academic performance patterns and relationships between different student features.

The project provides a practical understanding of data analysis and machine learning and demonstrates how these techniques can support educational decision-making.

The system is intended to assist academic analysis and should not be used as the sole basis for decisions about students.

16. REFERENCES

1. UCI Machine Learning Repository, Student Performance Dataset. https://archive.ics.uci.edu/dataset/320/student+performance
2. Scikit-learn Documentation. https://scikit-learn.org/
3. Pandas Documentation. https://pandas.pydata.org/docs/
4. NumPy Documentation. https://numpy.org/doc/
5. Matplotlib Documentation. https://matplotlib.org/
6. Seaborn Documentation. https://seaborn.pydata.org/